

Changyu (Thomason) Zhao

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EDUCATION

Stanford University

M.S. in Computer Science (Computer and Network Security)

GPA: **4.0/4.0**

Sep. 2025 – Jun. 2027 (expected)

Stanford, California

University of Wisconsin-Madison

B.S. in Computer Science (Honors) and Mathematics

GPA: **3.95/4.0**, David Dewitt Undergraduate Scholarship, Dean's List (all semesters)

Sep. 2021 – May 2025

Madison, Wisconsin

EXPERIENCES

Tesla

Jun. 2023 – Sep. 2023

Product Security Engineer Intern

Palo Alto, California

- Built an **agentic vulnerability research framework** integrating source, binary, and production build analysis, uncovering 10 previously unknown vulnerabilities including memory-corruption flaws in security-critical userspace and kernel components.
- Conducted **vulnerability research** across vehicle infotainment, connectivity, and 4 energy-product platforms, assessing exploitability and root causes while driving remediation and pre-release fix verification with product engineering teams.
- Hardened TCU firmware through kernel security configurations and systemd runtime protections, reducing the effective remotely exploitable attack surface by **~30%** and driving adoption of CFI enforcement for vehicle connectivity software.

UW-Madison MadS&P Lab

Sep. 2024 – Aug. 2025

Research Assistant, Advisor: Prof. Patrick McDaniel

Madison, Wisconsin

- Conducted **comprehensive evaluation** of LibIHT against Intel Pin and DynamoRIO on real-world workloads, demonstrating **150x speedup**, **99%+ code coverage**, and **complete resistance** to anti-debugging techniques employed by malware.
- Established **reproducible benchmarking infrastructure** with 21 coreutils binaries and 10 malware specimens, automating performance metrics collection and visualization in **Python**. Paper accepted by **ACM CCS 2025 SURE** workshop.

ASU SEFCOM Lab

May. 2024 – Sep. 2024

Research Assistant, Advisor: Prof. Yan Shoshitaishvili and Ruoyu Wang

Tempe, Arizona

- Co-designed Oxidizer, the first **Rust-focused decompiler** built atop angr, implementing control-flow recovery, function prototype inference, and macro inference to address the unique challenges of analyzing memory-safe compiled binaries.
- Developed **static type recovery engine** to extract Rust-specific type metadata from compiled binaries and integrates findings into decompilation, improving accuracy of data structure and function signature reconstruction for security analysis.

Tencent Xuanwu Lab

Jul. 2023 – Aug. 2023

Security Researcher Intern

Beijing, China

- Architected **LibIHT**, a cross-platform dynamic binary tracing framework in **C/C++/Python** that leverages Intel LBR/BTS hardware features for control-flow tracing, achieving **150x speedup** over Intel Pin on large-scale program analysis.
- Designed unified **kernel driver architecture** for Linux and Windows with abstraction layers for OS and hardware, reducing code duplication by **40%** while standardizing trace session management and improving cross-platform maintainability.
- Implemented **user-space API library** and debugger plugin system for **GDB/WinDbg** integration, enabling security analysts to visualize execution traces, identify anomalous control-flow patterns, and accelerate vulnerability discovery workflows.

ASU SEFCOM Lab

May. 2023 – Jul. 2023

Research Assistant, Advisor: Prof. Yan Shoshitaishvili

Tempe, Arizona

- Developed **LibHopper**, an automated exploitation primitive discovery tool combining debugger instrumentation with **symbolic execution** to identify exploitable memory corruption opportunities in library state structures.
- Constructed **exploitation primitives dataset** spanning critical C libraries (OpenSSL, OpenSSH, libcurl, libpng), providing structured risk assessments to inform security research in automated exploit generation and vulnerability scanning tooling.

PRESENTATIONS & PUBLICATIONS

Yibo Liu, ..., **Changyu Zhao**, ... "Oxidizer: Toward Concise and High-fidelity Rust Decompilation". *2026 IEEE Symposium on Security and Privacy (IEEE S&P)*. May 2026.

Changyu Zhao, ... "LibIHT: A Hardware-Based Approach to Efficient and Evasion-Resilient Dynamic Binary Analysis".

Proceedings of the 1st Workshop on Software Understanding and Reverse Engineering (SURE 2025). ACM, October 2025.

Black Hat Arsenal, *LIBIHT: A Cross-Platform Library for Accessing Intel Hardware Trace Features*, Las Vegas, NV, Aug. 2024

EXTRACURRICULAR ACTIVITIES

r3kapig CTF Team

Sep. 2021 – Present

Team Member

International

- Entered **DEFCON CTF world final** from 2022 to 2025, ranked 12th, 7th, 4th, and 5th respectively.
- Competed in **120+ CTF competitions**, expert in kernel and user-space exploitation and reverse engineering.

SKILLS

Languages: C/C++, Python, Rust, Go, Java, x86/x64, ARM, MIPS Assembly

Systems & Reverse Engineering: QEMU, GDB, WinDbg, IDA Pro, Ghidra

Security Analysis: Angr, AFL++, Frida, Joern, Wireshark, Metasploit, Kali